REPORT ON MY RESEARCH EXPERIENCE

Global Research & Standards — Member 3

Assignment Area: Global WAN Technologies, Wireless Technologies and International Networking Standards

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Date: 27 August 2026

1. Introduction

As Member 3 of the group, my main responsibility was to provide the global perspective for our networking research. My research focused on understanding how WAN and wireless technologies are used internationally and how modern networks follow global technical standards. This included researching global WAN technologies, wireless technologies, IEEE 802.11, 3GPP, ITU-T, 5G Standalone, LEO satellites, Starlink and global networking trends.

2. My Research Experience

The research experience was useful because it allowed me to move beyond basic definitions of networking technologies and understand how they are applied in real-world international environments. I learned that global networking is not based on one technology. Organizations normally combine different technologies depending on location, cost, availability, performance and reliability requirements.

One of the most interesting parts of the research was learning about LEO satellite networks. Before this research, I mainly associated satellite communication with high latency and remote connectivity. Researching LEO systems showed me how newer satellite constellations can provide lower latency and high-speed connectivity, making them useful for businesses, ships, aircraft, remote communities and backup WAN connections.

3. What I Learned About Global WAN Technologies

I learned that technologies such as MPLS, SD-WAN, VPNs, leased lines, 4G, 5G and satellite connectivity can all play different roles in a global WAN. SD-WAN was particularly important because it can use multiple types of connections and intelligently route traffic according to application and network conditions.

This research helped me understand the importance of redundancy. A multinational organization cannot always depend on a single network connection because fibre cuts, equipment failures, congestion or geographic limitations can interrupt communication. Using multiple connectivity options can therefore improve availability and business continuity.

4. What I Learned About Wireless Standards

Researching IEEE 802.11 helped me understand that Wi-Fi is not simply a product or brand name. It is based on a family of international technical standards that define how wireless LAN devices communicate. I also learned that standards continue to evolve as users require higher performance, greater efficiency, better security and support for new applications.

I also learned about 3GPP and its role in developing specifications for cellular technologies. Understanding the relationship between 4G, 5G Non-Standalone and 5G Standalone helped me see how mobile networks evolve while maintaining interoperability between equipment and service providers.

5. Understanding ITU-T and Global Standards

The research introduced me to the role of ITU-T in international telecommunications standardization. I learned that global networking requires cooperation between countries, telecommunications operators, equipment manufacturers and standards organizations. International recommendations

help create common technical expectations and support interoperability.

This was important to me because it showed that networking is not only about configuring routers, switches and access points. At a global level, networking also involves standards, regulation, spectrum, interoperability, security and international coordination.

6. Multinational Case Study Experience

The Angelicoussis Group case study made the research more practical. The organization operates a global maritime fleet, where ships can travel far beyond the reach of terrestrial networks. The use of Starlink's LEO satellite connectivity demonstrated how modern WAN technology can solve a real business problem.

Studying this case helped me understand the connection between theory and practice. Concepts such as bandwidth, latency, availability and remote connectivity are not only classroom topics; they directly affect how multinational organizations operate, communicate and manage their services.

7. Challenges I Experienced

One challenge during the research was dealing with a large amount of technical information. Different technologies use specialized terminology, and it was necessary to compare information from standards organizations, technology providers and industry reports. Another challenge was separating general marketing claims from technical information that could be used reliably in an academic report.

I also had to understand how different standards organizations relate to one another. IEEE, 3GPP and ITU have different roles, and understanding those roles was important when comparing technologies against global standards.

8. Skills I Developed

Through this research, I developed several useful skills. These include online research, source evaluation, technical reading, comparison of networking technologies, academic report writing and referencing. I also improved my ability to explain complex networking concepts in a simpler way.

The research also strengthened my understanding of how theoretical networking concepts can be connected to real organizational requirements. This is valuable for my future work in IT because technical professionals must be able to research technologies before recommending or implementing them.

9. Reflection

Overall, this research was a valuable learning experience. It gave me a broader understanding of networking beyond a local network environment. I learned that global connectivity depends on a combination of technologies and internationally accepted standards. I also gained a better appreciation for the importance of reliable sources when conducting technical research.

The research changed my understanding of modern networking because I now see WAN, wireless, mobile and satellite technologies as interconnected parts of a larger global communication system. The knowledge gained from this task will help me approach future networking projects with a more global and standards-based perspective.

10. Conclusion

My role as the Global Research & Standards Lead allowed me to investigate technologies that are shaping modern international networking. Researching WAN technologies, IEEE 802.11, 3GPP, ITU-T, 5G Standalone, LEO satellites and Starlink improved both my technical knowledge and my research skills. The multinational case study helped me understand how these technologies can solve real-world problems. Overall, the experience improved my ability to conduct technical research, evaluate information and connect networking theory with practical global applications.